DALHOUSIE UNIVERSITY

Faculty of Engineering

Department of Electrical and Computer Engineering

ECED3901 — Electrical Engineering Design II

Detailed System Operation and Architecture Diagram Report

Prepared by

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Presented to

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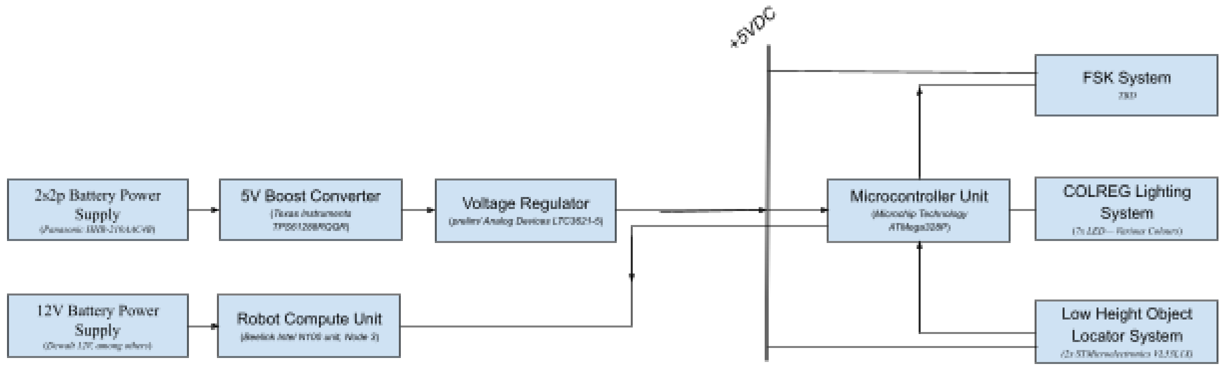
February 4, 2026

Hardware Subsystems

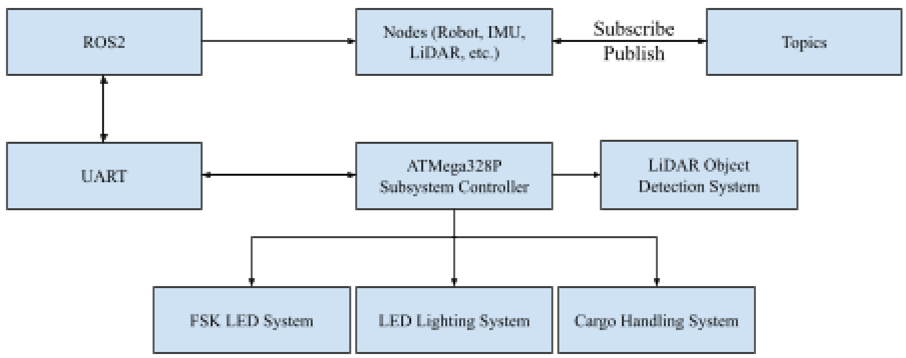


Team Three — Hardware Subsystem Architectural Diagram

Hardware Component Descriptions/TRL's	***TRL***	***TRR[1]***
• 2.4V Output – 2s2p Battery Power Supply (*Panasonic HHR-210AAC4B, 1.2V*)	**9**	9
○ This will be the primary power source for Team 3's auxiliary system. While not fancy, these rechargeable units from Panasonic are renowned for their high quality, are dependable, and can be easily replaced for continued testing. Team 3 will have multiple sets on hand, and based on testing, estimates this solution will provide up to 4 hours of auxiliary system power.		
• 5V Boost Converter (*Texas Instruments TPS61288RQQR*)	**4**	6
○ While the TPS61288RQQR is an extremely reliable, and efficient unit (*92.4% efficiency measured in testing*), thus far, Team 3 has only proven results in LTSpice simulations, successfully boosting voltage to 7.5V at 500mA with minimal output ripple. This will be regulated down to 5V by Team 3's Voltage regulator.		
• Voltage Regulator (*prelim/ Analog Devices LTC3621-5)*	**2**	4
○ Similar to the TPS61288RQQR, the LTC3621-5 is a widely proven voltage regulator IC. Team 3 will be using this IC for regulating down the dropping transient 7.5V to 5V. This gives us some buffer, as the LTC3621-5 requires a 5V input, allowing the system to draw from the battery source longer. The TRL here is low, as Team 3 has not yet locked down the LTC3621-5. This IC is subject to change.		

- +5VDC Bus 5 7
  - Being on the output end of the Voltage Regulator, the +5V DC will be the primary power bus for all output auxiliary systems. All subsystems currently require 5V DC of some sort at the moment. This will be trivial to implement, but our TRL remains low as the Voltage Regulator and Boost Converter are yet to be implemented in practice.
- Microcontroller Unit (*Microchip Technology ATMega328P*) 8 8
  - Team 3 will be implementing an ATMega328P as our subsystem controller, as they are cheap, widely available, and most importantly, Team 3 members already have a strong familiarity with the MCU's ecosystem.
- Lighting System (*7x LED— Various Colours*) 4 5
  - Team 3's COLREG requirement will be fulfilled by a series of LED's, controlled by the ATMega328P. Currently, Team 3 is planning on using set colour LED's, controlled by hardware timers onboard the MCU. Multiplexing is not currently planned, but can be implemented if pinout limitations require.
- Cargo Handling System 2 3
  - Team 3's Cargo Handling system is very early in development, however, ideation and testing is currently taking place. Team 3 is aiming to create a hardware free cargo handling system that prioritizes simplicity, and straying away from motors, lift systems, and electromagnets. They will be used as backup after physical navigation testing takes place.
- FSK System 1 2
  - Very early in development. Currently waiting for more course information before continuing development.
- Low Height Object Locator System, (*2x STMicroelectronics VL53L1X*) 1 1
  - Again, very early in development. Team 3 will be implementing two low height VL53L1X 1-D LiDAR systems to assist in ground object location and tracking. This will interact with ROS to stream information into the object probability map.

Software/Firmware Tree



Team Three — Software Subsystem Architectural Diagram

Software Component Descriptions/TRL's	***TRL***	***TRR[1]***
● ROS2	**8**	4
○ This will be the middleware used to allow the robot to function. It works by using nodes, these nodes will then subscribe or publish to topics. By subscribing to a topic, the node is able to read the information that is published to the topic. Nodes will be used to communicate information to the MCU through USART.		
● Lighting System	**4**	3
○ The team will use the ATMega328P MCU connected with several LEDs in order to implement a COLREG system on the robot. Team 3 will program this system using C code and multiplex the LEDs.		
● FSK LED System	**1**	1
○ This program will be uploaded to the ATMega328P MCU and will be used to control the FSK system.		
● LiDAR Objection Detection System	**2**	2
○ The use of a 1D LiDAR component to detect the cargo and raft will be the primary way for the robot to detect cargo. The software compares the distance from the spinning LiDAR on top, with the fixed 1D LiDAR that is located on the lower half of the robot. The robot detects an object if a discrepancy in the two LiDAR distances occurs.		

TRR[1] – Testing and Research Remaining, Ranked (*One being highest priority*)

Question 2:

Odometry serves as a localization technique, enabling the robot to estimate its instantaneous position and orientation (*pose*) relative to a starting reference point. By integrating real-time data from on-board IMU sensors among others, ROS calculates incremental displacements to track movement across a surface, in Team 3's use case, assisting with course navigation.

Implementing a precise odometry model is foundationally vital to Team 3's navigation strategy. It provides the base spatial awareness necessary to autonomously traverse the course, execute precise manoeuvres, and reliably manage cargo pick-up and delivery navigation as specified in the client requirements.

Error is the difference between a measured value and the true value, whereas uncertainty is the estimated range of doubt as to where the true value lies. Error is the deviation between a measured value and the true value, while uncertainty represents the estimated range of potential doubt surrounding the true value.

Furthermore, a systemic error is an error which occurs consistently due to faults in the hardware or software. An example could be the robot veering left due to one of the motors being weaker. A random error, on the other hand, is an error which occurs inconsistently by chance due to outside random factors. An example could be the robot sometimes veering left or right due to small slips on the surface or environmental factors.

There are several factors that contribute to the uncertainty of the robot. Some systemic errors which could happen are hardware errors. For example, the motors may not spin at the same rate or the LiDAR may be damaged in some way. As for random errors, the lab's floor is quite dirty and over the week, students have been walking in the course and tracking the dirt. This may lead to the robot slipping and small errors accumulating. These small errors accumulate over time and increase the uncertainty. It is necessary to take in information with the lidar and constantly correct this uncertainty. The Gazebo simulator does not capture these factors, which can make it difficult to accurately simulate the robot.

Question 3:

The robot's perception system is categorized by three distinct sensor types, each serving a specific role in feedback and environmental awareness:

- **Proprioceptive Sensors**: These provide the robot with internal feedback regarding its own physical state and kinematics. For Team 3's design, this primarily involves measuring joint angles and motor velocities via wheel encoders. This data is needed for odometry calculations used to estimate relative displacement. Some example use cases of proprioceptive sensors are IMUs, encoders (*both linear and rotational*), and potentiometers.

- **Exteroceptive Sensors**: These capture information from the robot's external environment. By using sensors that measure distance (*e.g. LiDAR, Ultrasonic*), light intensity, or acoustic signals, the robot can detect obstacles and recognize external landmarks necessary for global localization and mapping. Use cases are seen in LiDAR, Ultrasonic Sensors, and Cameras.

- **Interoceptive Sensors**: While proprioceptive sensors focus on motion, interoceptive sensors monitor the internal status of a system. These are used to track parameters such as battery voltage, current draw, and component temperature to ensure the robot operates within safe electrical and thermal limits. In-practice examples include thermocouples, pressure transducers, and voltage sensors.

Team 3 has currently implemented onboard LiDAR and IMU systems. Additional 1D LiDAR sensors are being considered to enhance the cargo retrieval system's tracking accuracy. Team 3 believes that the current set of proprioceptive and exteroceptive sensors is considered sufficient for the project requirements, however are considering additional near range sensors.

Interoceptive sensors, which measure internal conditions, have been deemed not necessary.

Question 4:

The encoder system provides data on the Proprioceptive elements of the robot, such as the angular position, rotational angle, and the direction of translation. This allows Team 3 to navigate the course and move properly to retrieve the cargo.

The 25SG-370CA-EN gear motor has an incremental quadrature encoder with A and B channels. The encoder operates at 3.3 ~ 5 V and connects to a microcontroller, allowing accurate odometry for the robot. In the incremental quadrature encoder, Channel A pulses a square wave at increments of motion and channel B is 90° out of phase of channel A to tell the direction of rotation.

LiDAR will prove to be essential for obstacle detection and navigation within the course. Currently, the design implements a high-mounted YDLiDAR for 360-degree scanning, needed for navigational probability matrix mapping, and will include multiple 1D VL53L1X sensors. The YDLiDAR is much more resilient to ambient lighting, which results in a clearer and more accurate signal.

The 1D VL53L1X sensors are planned to be used to determine object location, required for Team 3's planned cargo retrieval system. The VL53L1X appears to offer reliable detection up to 4 meters with a programmable Region of Interest (ROI).

In brief,

- *Triangulation* can make a determination of distances by measuring the angle of reflected light. A laser hits an object, and a sensor looks at it from an offset angle. As the object moves closer or farther, the angle of the reflected light changes.
- *Time-of-Flight* makes a determination of distance using raw speed. A light pulse is fired, and a timer tracks how long it takes to bounce back.

- *Phase-Shifting* relies on wave patterns. It sends a continuous, traditionally sine wave light beam, and compares the peak/valley alignment of the sent wave compared to the returning wave. The shift observed in the wave is where the distance is measured. Interestingly, however not required, phase-shifting if correctly implemented can relay material property information back to an operator (*or in our case, robot*). This is done through observation of the material's refractiveness, and on high resolution systems, visible surface roughness.

To Team 3's current knowledge, the high mounted YDLiDAR system is built on a time-of-flight system. Team 3 plans to implement a similar time-of-flight system for the 1D VL53L1X sensors, solely due to design simplicity and deliverable requirements.

Client requirements state that the robot must autonomously navigate the course and retrieve the cargo from the port while avoiding physical contact with obstacles in its path. Team 3 will be using a 2D high mounted LiDAR to map the environment and build an accurate plan for autonomous navigation using SLAM (Simultaneous Localization and Mapping). Additionally, the team has decided to add low mounted LiDAR to assist in detection of low-profile hazards/objects, increasing pathing matrix precision.

Together, both LiDAR systems address several key deliverable requirements, as detailed below:

- **2D High-Mounted LiDAR (Positioning & Navigation):**

  The 2D high YDLiDAR, situated on top of the robot, will be essential in autonomous navigation. Other than the significance mentioned above, the route pathing required for the coastal path navigation will be almost entirely handled by the high-mounted sensor, comparing to a local map.

- **Low-Mounted LiDAR (Object Detection):**

  The low mounted 1D LiDAR will be used to detect the translational movement of threats in the robot's path. The pirates are expected to return a LiDAR signal at a height of 3 to 12 inches off the ground. Team 3 will display threat level indicators based on radius — for which the low LiDAR will provide the precise distance data needed to trigger these specific LED states. Avoiding contact with low-profile hazards such as cargo, other vessels, or overboard persons will hopefully prove to be much easier with such a system.

Needed Systems (*Ongoing Development*):

1. Optical Communication System (For Pirate Deterrence)

2. Cargo Handling & Rescue System

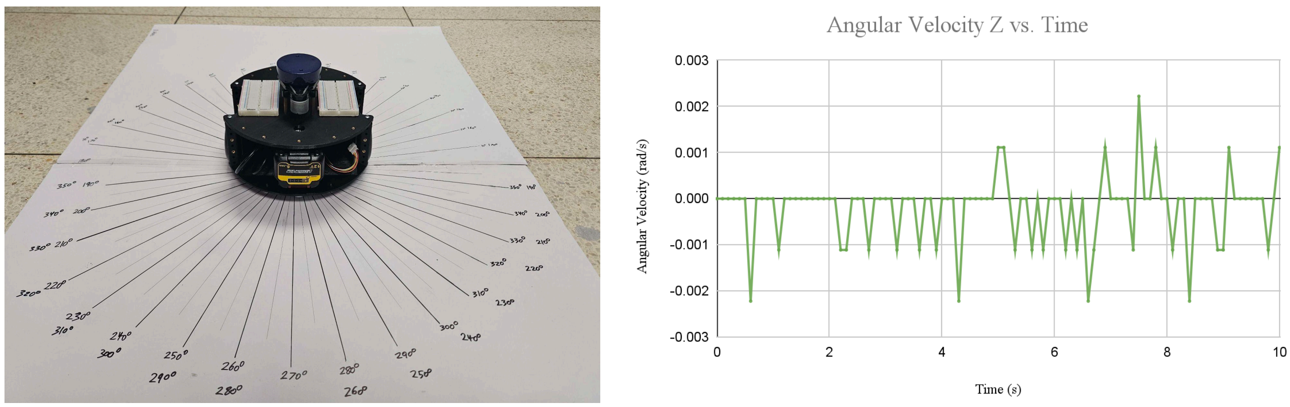
3. Lighting Safety System (COLREGS)

4. Close-Range Sensors

While LiDAR is great for mapping, a common pitfall proves to be blind spots at very close ranges (*<10-20 cm*). This will need to be further investigated by Team 3, but is noted in the VL53L1X datasheet as 4 cm.

Question 5:

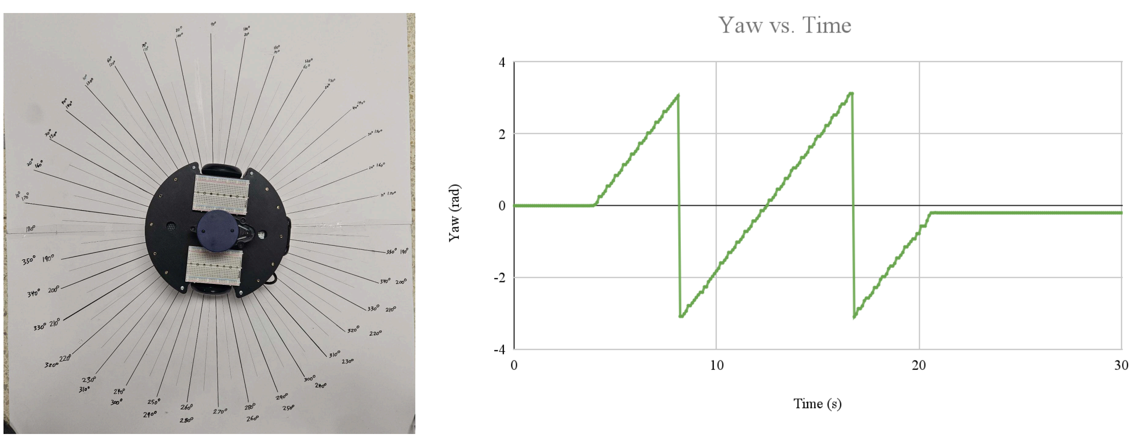
A template for angular observation was created, which can be seen below. The template contains several angular values and long lines for ease of measurement. Using the IMU data stream, the following data was obtained, and can be displayed for the angular velocity:



Team Three — Angular Jitter Observation Testing

Notably, the IMU reads non-zero angular velocity even while the robot is completely static, with a maximum error of about $0.0022\ \mathrm{rad/s}$. According to the datasheet, the maximum output noise is about $0.3\ \mathrm{degrees/s}$ or $\sim 0.0052\ \mathrm{rad/s}$. Our observation comfortably sits under this threshold.

Next, the IMU reported yaw values while rotating the robot in place using the`teleop` executable. After completing roughly two revolutions, the robot came to rest at an angle of about 353 degrees (*pictured below*).



Team Three — Yaw Observation Testing

The measured yaw initializes at zero and exhibits a standard wrap-around discontinuity at π radians as the robot approaches its second full revolution. The final recorded value of -0.198407 radians corresponds to 348.63°on the test template.

A comparison between the ground truth and the IMU-derived orientation reveals a cumulative heading error of approximately 5° over the two-revolution interval. This discrepancy is likely attributable to high-frequency sensor noise and stochastic drift exacerbated by rapid accelerations. While a 5° margin of error is non-negligible, it remains within the tolerable threshold for the requirements of this challenge. To enhance localization accuracy in future iterations, this error can be mitigated through sensor fusion, integrating additional proprioceptive data from wheel encoders with exteroceptive LiDAR measurements via an Extended Kalman Filter (EKF) or similar estimation algorithm.